

E-Learning Portal Backend System

Using Java Spring Boot and MongoDB

Project Report

Project Title:	E-Learning Portal Backend System
Technology:	Java Spring Boot, MongoDB
Database:	MongoDB Community Server
API Testing Tool:	Postman
Documentation Tool:	Swagger OpenAPI

Submitted By

Prathviraj Achut Chavan
24BCE10116
CSE [CORE]
VIT Bhopal University

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Contents

1	Abstract	3
2	Objectives	3
3	Expected Outcome	3
4	Technologies Used	4
5	Platform Used	4
6	Backend Language Used	4
7	System Architecture	5
8	Database Collections	5
8.1	Students	5
8.2	Instructors	5
8.3	Courses	6
8.4	Lessons	6
8.5	Enrollments	6
8.6	Reminders	6
9	Entity Relationships	7
9.1	Instructor to Courses	7
9.2	Course to Lessons	7
9.3	Student to Course	7
10	Modules Implemented	7
10.1	Student Management	7
10.2	Instructor Management	7
10.3	Course Management	7
10.4	Lesson Management	8
10.5	Enrollment Management	8
10.6	Reminder System	8
11	Advanced MongoDB Features	8
11.1	Student Dashboard API	8
11.2	Course Search API	8
11.3	Course Statistics API	9
11.4	Course Completion Tracker	9
12	REST APIs Developed	9
12.1	Student APIs	9
12.2	Instructor APIs	9
12.3	Course APIs	9
12.4	Lesson APIs	10
12.5	Enrollment APIs	10

12.6 Reminder APIs	10
13 Output Screenshots	11
13.1 MongoDB Database Setup	11
13.2 Database Schema Design	12
13.3 Student Management APIs	13
13.4 All Students Collection	14
13.5 Instructor Management APIs	15
13.6 Course Management APIs	16
13.7 Lesson Management APIs	17
13.8 Enrollment Management APIs	18
13.9 Reminder Management APIs	19
13.10 Student Dashboard API	20
13.11 Course Statistics API	20
13.12 Course Completion Tracker	21
14 Testing	21
15 Summary	21
16 Conclusion	22

1 Abstract

The E-Learning Portal Backend System is developed using Java Spring Boot and MongoDB to manage online learning activities. The system provides functionalities for managing students, instructors, courses, lessons, enrollments, reminders, and course completion tracking. It demonstrates CRUD operations, MongoDB relationships, aggregation framework, RESTful APIs, and API testing using Postman and Swagger UI.

The project follows a backend-only architecture and provides secure and scalable data management using MongoDB collections and Spring Boot REST APIs.

2 Objectives

- To develop a backend system for an online learning platform.
- To implement CRUD operations using MongoDB.
- To establish relationships between entities using MongoDB references.
- To provide RESTful APIs for data management.
- To implement aggregation operations such as Dashboard, Statistics, and Completion Reports.
- To demonstrate practical usage of Java Spring Boot with MongoDB.

3 Expected Outcome

The system should be capable of:

- Managing students and instructors.
- Managing courses and lessons.
- Enrolling students into courses.
- Tracking course completion progress.
- Managing student reminders.
- Searching courses efficiently.
- Generating course statistics and reports.
- Providing API documentation using Swagger.

4 Technologies Used

Technology	Version
Java	17
Spring Boot	3.x
Maven	3.9+
MongoDB Community Server	8.x
MongoDB Java Driver	Latest Compatible
Swagger OpenAPI	2.8.9
Postman	Latest
Visual Studio Code	Latest
Git	Latest

Table 1: Technologies Used

5 Platform Used

- Operating System: Windows 10/11
- IDE: Visual Studio Code
- Database: MongoDB
- API Testing Tool: Postman
- API Documentation Tool: Swagger UI

6 Backend Language Used

Java (Spring Boot Framework)

Java Spring Boot is used to create REST APIs and connect with MongoDB for performing CRUD operations and aggregation operations.

7 System Architecture

Student
|
Enrollment
|
Course
|
Lesson

Instructor
|
Course

Reminder
|
Student

8 Database Collections

8.1 Students

Stores student information.

Fields:

- studentId
- name
- email
- phone

8.2 Instructors

Stores instructor information.

Fields:

- instructorId
- name
- email
- specialization

8.3 Courses

Stores course details.

Fields:

- courseId
- courseName
- description
- duration
- instructorId

8.4 Lessons

Stores lesson information.

Fields:

- lessonId
- courseId
- title
- videoUrl

8.5 Enrollments

Stores student-course mapping.

Fields:

- enrollmentId
- studentId
- courseId
- progress

8.6 Reminders

Fields:

- reminderId
- studentId
- courseId
- message
- reminderDate
- status

9 Entity Relationships

9.1 Instructor to Courses

Relationship Type: One-to-Many

One instructor can teach multiple courses.

9.2 Course to Lessons

Relationship Type: One-to-Many

One course can contain multiple lessons.

9.3 Student to Course

Relationship Type: Many-to-Many

Implemented through Enrollment collection.

10 Modules Implemented

10.1 Student Management

- Add Student
- View Students
- Update Student
- Delete Student

10.2 Instructor Management

- Add Instructor
- View Instructor
- Update Instructor
- Delete Instructor

10.3 Course Management

- Create Course
- Update Course
- Delete Course
- Search Course

10.4 Lesson Management

- Add Lesson
- View Lesson
- Update Lesson
- Delete Lesson

10.5 Enrollment Management

- Enroll Student
- View Enrollment
- Update Progress
- Delete Enrollment

10.6 Reminder System

- Create Reminder
- View Reminders
- Update Reminder Status
- Delete Reminder

Purpose: Helps students track pending tasks and course deadlines.

11 Advanced MongoDB Features

11.1 Student Dashboard API

Uses:

- `$match`
- `$lookup`

Purpose: Displays all courses enrolled by a student along with course information.

11.2 Course Search API

Searches courses using course name.

11.3 Course Statistics API

Uses:

- \$group
- \$sum

Purpose: Displays total students enrolled in each course.

11.4 Course Completion Tracker

Uses:

- \$group
- \$avg
- \$lookup

Purpose: Calculates average completion percentage of every course.

12 REST APIs Developed

12.1 Student APIs

```
POST    /students
GET     /students
GET     /students/{id}
PUT     /students/{id}
DELETE  /students/{id}
```

12.2 Instructor APIs

```
POST    /instructors
GET     /instructors
GET     /instructors/{id}
PUT     /instructors/{id}
DELETE  /instructors/{id}
```

12.3 Course APIs

```
POST    /courses
GET     /courses
GET     /courses/{id}
PUT     /courses/{id}
DELETE  /courses/{id}
GET     /courses/search/{name}
```

12.4 Lesson APIs

POST /lessons
GET /lessons
GET /lessons/{id}
PUT /lessons/{id}
DELETE /lessons/{id}

12.5 Enrollment APIs

POST /enrollments
GET /enrollments
GET /enrollments/{id}
PUT /enrollments/{id}
DELETE /enrollments/{id}
GET /enrollments/dashboard/{studentId}
GET /enrollments/statistics

12.6 Reminder APIs

POST /reminders
GET /reminders
GET /reminders/student/{studentId}
PUT /reminders/{id}
DELETE /reminders/{id}

13 Output Screenshots

13.1 MongoDB Database Setup

```
ElearningDB> db.reminders.insertMany([
|
| {
|
|   reminderId:"R201",
|
|   studentId:"ST201",
|
|   courseId:"C101",
|
|   message:"Complete Java Assignment",
|
|   reminderDate:"2026-06-20",
|
|   status:"Pending"
|
| },
|
| {
|
|   reminderId:"R102",
|
|   studentId:"ST102",
|
|   courseId:"C102",
|
|   message:"Watch Spring Boot Lesson",
|
|   reminderDate:"2026-06-21",
|
|   status:"Pending"
```

13.2 Database Schema Design

Student ^ Collapse all object
studentId string
name string
email string
phone string

Reminder ^ Collapse all object
reminderId string
studentId string
courseId string
message string
reminderDate string
status string

Lesson ^ Collapse all object
lessonId string
courseId string
title string
videoUrl string

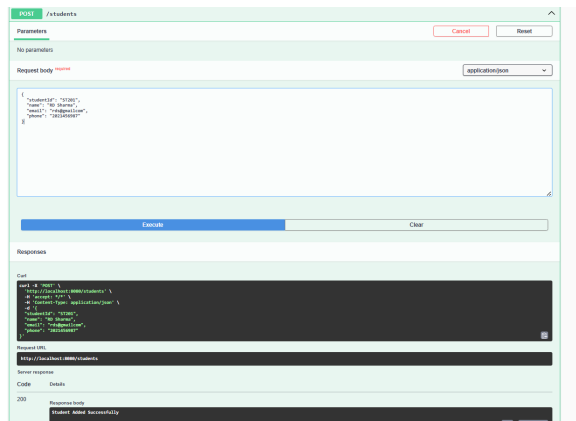
Instructor ^ Collapse all object
instructorId string
name string
email string
specialization string

Enrollment ^ Collapse all object
enrollmentId string
studentId string
courseId string
progress integer int32

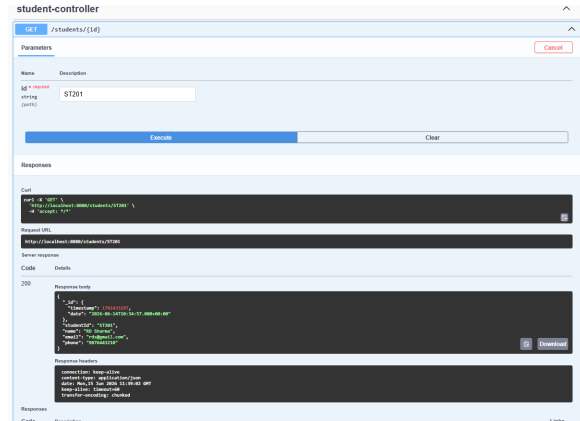
Course ^ Collapse all object
courseId string
courseName string
description string
duration string
instructorId string

Figure 2: MongoDB Collections and Relationships Schema

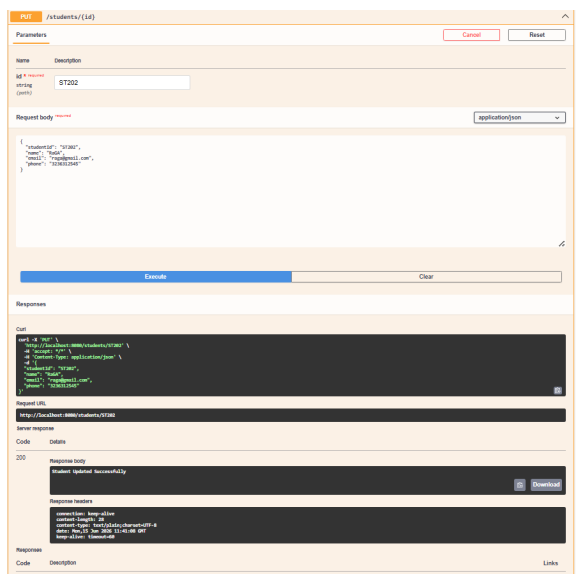
13.3 Student Management APIs



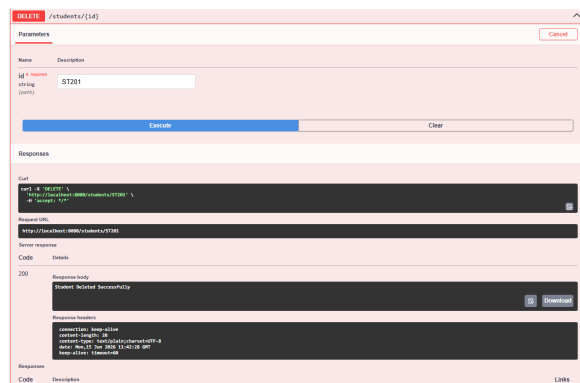
(a) Create Student



(b) Retrieve Student



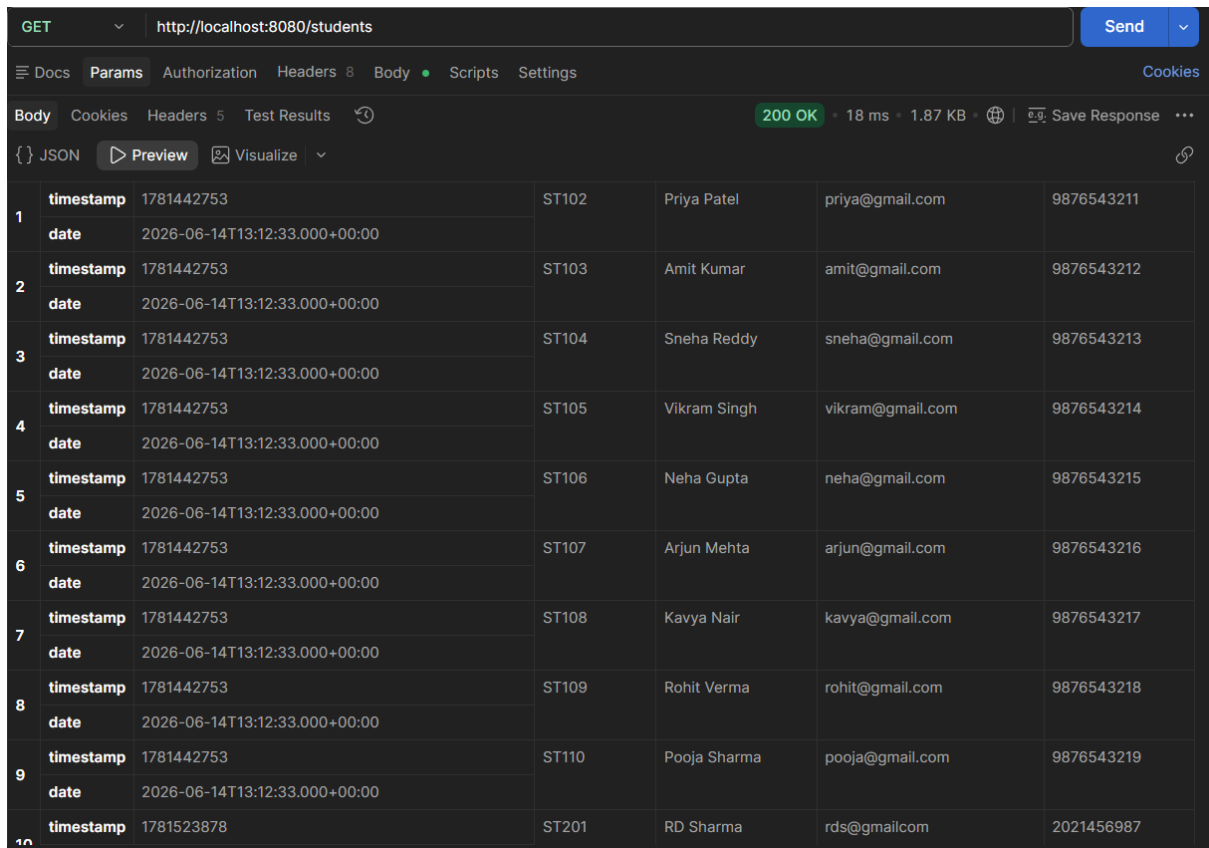
(c) Update Student



(d) Delete Student

Figure 3: Student CRUD Operations

13.4 All Students Collection



GET <http://localhost:8080/students> Send

Docs Params Authorization Headers 8 Body • Scripts Settings Cookies

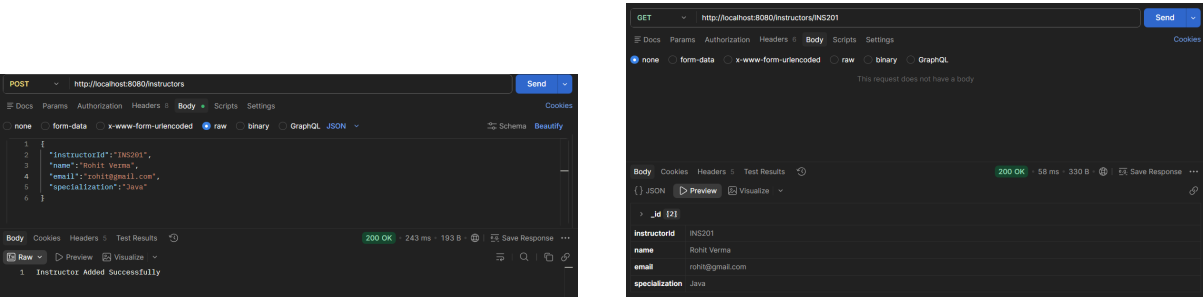
Body Cookies Headers 5 Test Results 200 OK • 18 ms • 1.87 KB Save Response

{ } JSON Preview Visualize

1	timestamp	1781442753	ST102	Priya Patel	priya@gmail.com	9876543211
	date	2026-06-14T13:12:33.000+00:00				
2	timestamp	1781442753	ST103	Amit Kumar	amit@gmail.com	9876543212
	date	2026-06-14T13:12:33.000+00:00				
3	timestamp	1781442753	ST104	Sneha Reddy	sneha@gmail.com	9876543213
	date	2026-06-14T13:12:33.000+00:00				
4	timestamp	1781442753	ST105	Vikram Singh	vikram@gmail.com	9876543214
	date	2026-06-14T13:12:33.000+00:00				
5	timestamp	1781442753	ST106	Neha Gupta	neha@gmail.com	9876543215
	date	2026-06-14T13:12:33.000+00:00				
6	timestamp	1781442753	ST107	Arjun Mehta	arjun@gmail.com	9876543216
	date	2026-06-14T13:12:33.000+00:00				
7	timestamp	1781442753	ST108	Kavya Nair	kavya@gmail.com	9876543217
	date	2026-06-14T13:12:33.000+00:00				
8	timestamp	1781442753	ST109	Rohit Verma	rohit@gmail.com	9876543218
	date	2026-06-14T13:12:33.000+00:00				
9	timestamp	1781442753	ST110	Pooja Sharma	pooja@gmail.com	9876543219
	date	2026-06-14T13:12:33.000+00:00				
10	timestamp	1781523878	ST201	RD Sharma	rds@gmail.com	2021456987

Figure 4: All Student Records Stored in MongoDB

13.5 Instructor Management APIs



(a) Create Instructor

(b) Retrieve Instructor

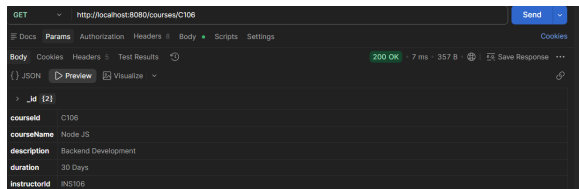
The screenshot shows a GET request to 'http://localhost:8080/instructors' returning a 200 OK status. The response is a JSON array of instructor objects, each containing an id, timestamp, date, instructorid, and name.

	_id	timestamp	date	instructorid	name
0	timestamp	1781348998	2026-06-13T11:09:58.000+00:00	INS101	Amit Verma
1	timestamp	1781442802	2026-06-14T13:13:22.000+00:00	INS102	Rakesh Gupta
2	timestamp	1781442802	2026-06-14T13:13:22.000+00:00	INS103	Suresh Kumar
3	timestamp	1781442802	2026-06-14T13:13:22.000+00:00	INS104	Anita Sharma
4	timestamp	1781442802	2026-06-14T13:13:22.000+00:00	INS105	Kiran Rao
5	timestamp	1781442802	2026-06-14T13:13:22.000+00:00	INS106	Rajesh Singh
6	timestamp	1781442802	2026-06-14T13:13:22.000+00:00	INS107	Meena Patel
7	timestamp	1781442802	2026-06-14T13:13:22.000+00:00	INS108	Arvind Nair
8	timestamp	1781442802	2026-06-14T13:13:22.000+00:00	INS109	Deepak Jain

(c) All Instructors

Figure 5: Instructor Management Operations

13.6 Course Management APIs



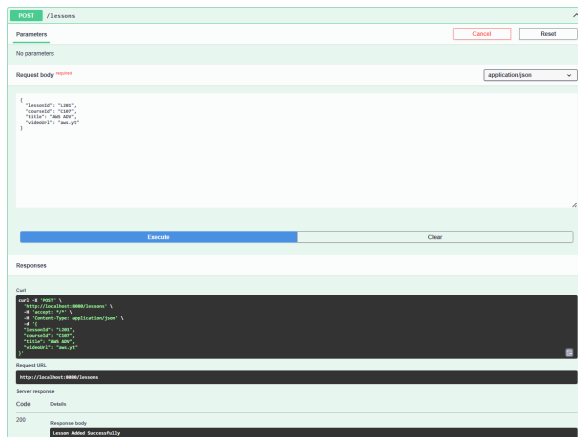
(a) Retrieve Course

	timestamp	courseid	courseName	description	duration	instructorid
0	17813489052	C101	Java Programming	Complete Java Course	30 Days	INS101
1	17813489052	C102	MongoDB	MongoDB Fundamentals	20 Days	INS102
2	17813489052	C103	Spring Boot	Spring Boot Development	25 Days	INS101
3	17813489052	C104	Python	Python for Beginners	40 Days	INS103
4	17813489052	C105	Web Development	HTML, CSS, JavaScript	35 Days	INS104
5	1781442954	C106	Node JS	Backend Development	30 Days	INS106
6	2028-06-14T13:15:54.000+00:00	C107	AWS Cloud	Cloud Computing	45 Days	INS107
7	1781442954	C108	DevOps	CI/CD and Docker	40 Days	INS108
8	1781442954	C109	Data Science	Analytics and Visualization	50 Days	INS109

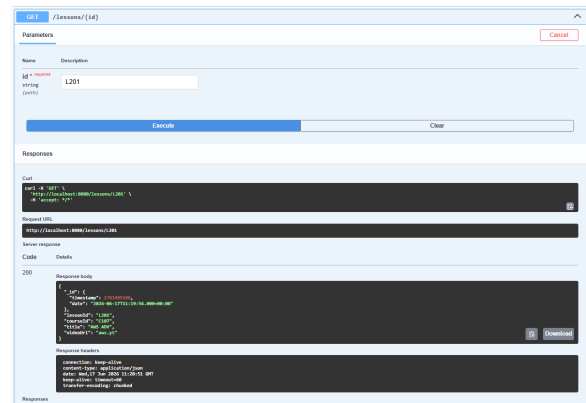
(b) All Courses

Figure 6: Course Management Operations

13.7 Lesson Management APIs



(a) Create Lesson



(b) Retrieve Lesson

GET http://localhost:8080/lessons

Send

Docs Params Authorization Headers 8 Body Scripts Settings Cookies

Query Params

Key	Value	Description	Bulk Edit
-----	-------	-------------	-----------

Body Cookies Headers 5 Test Results

200 OK · 28 ms · 1.02 KB · Save Response

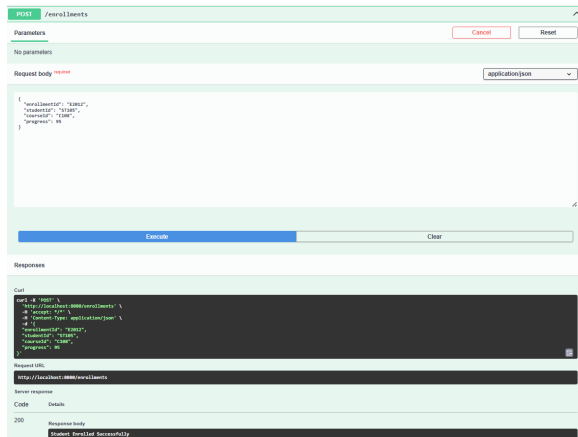
JSON Preview Visualize

	_id	timestamp	date	lessonId	courseId	lessonTitle	videoUrl
0		1781349212	2026-06-13T11:13:32.000+00:00	L101	C101	Introduction to Java	java_intro.mp4
1		1781349212	2026-06-13T11:13:32.000+00:00	L102	C101	Variables and Data Types	java_variables.mp4
2		1781349212	2026-06-13T11:13:32.000+00:00	L103	C101	Control Statements	java_control.mp4
3		1781349212	2026-06-13T11:13:32.000+00:00	L104	C102	MongoDB Basics	mongodb_basics.mp4
4		1781349212	2026-06-13T11:13:32.000+00:00	L105	C103	Spring Boot Introduction	springboot_intro.mp4

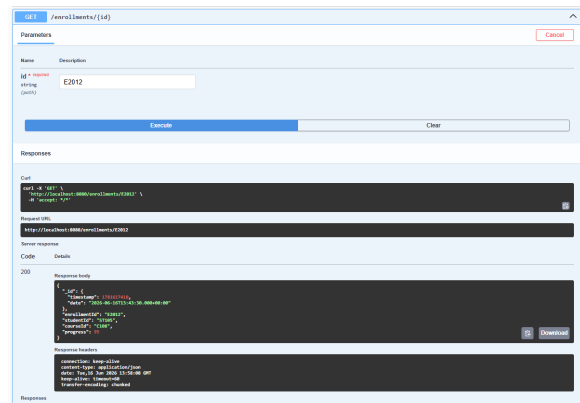
(c) All Lessons

Figure 7: Lesson Management Operations

13.8 Enrollment Management APIs



(a) Create Enrollment



(b) Retrieve Enrollment

GET http://localhost:8080/enrollments

200 OK · 18 ms · 1.3 KB · Save Response

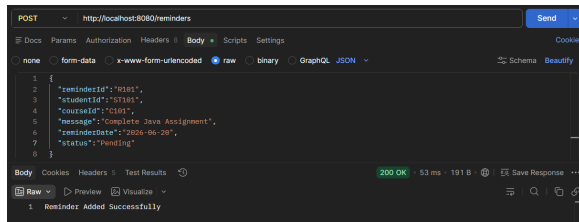
JSON Preview Visualize

	_id	timestamp	enrollmentid	studentid	courseid	progress
0	timestamp	1781349567	E101	ST101	C101	75
	date	2026-06-13T11:19:27.000+00:00				
1	timestamp	1781349567	E102	ST102	C102	60
	date	2026-06-13T11:19:27.000+00:00				
2	timestamp	1781349567	E103	ST103	C103	90
	date	2026-06-13T11:19:27.000+00:00				
3	timestamp	1781443186	E106	ST106	C106	45
	date	2026-06-14T13:19:46.000+00:00				
4	timestamp	1781443186	E107	ST107	C107	90
	date	2026-06-14T13:19:46.000+00:00				
5	timestamp	1781443186	E108	ST108	C108	25
	date	2026-06-14T13:19:46.000+00:00				
6	timestamp	1781443186	E109	ST109	C109	60
	date	2026-06-14T13:19:46.000+00:00				
7	timestamp	1781443186	E110	ST110	C110	80
	date	2026-06-14T13:19:46.000+00:00				

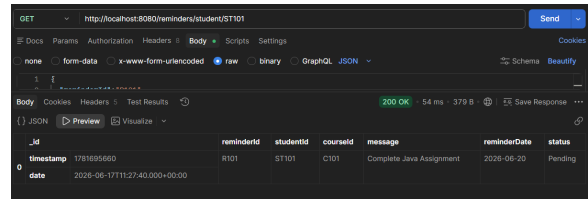
(c) All Enrollment Records

Figure 8: Enrollment Management Operations

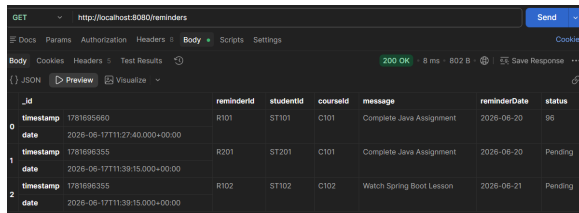
13.9 Reminder Management APIs



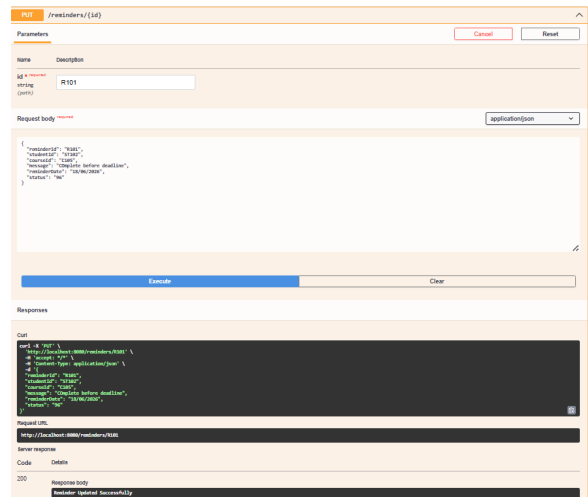
(a) Create Reminder



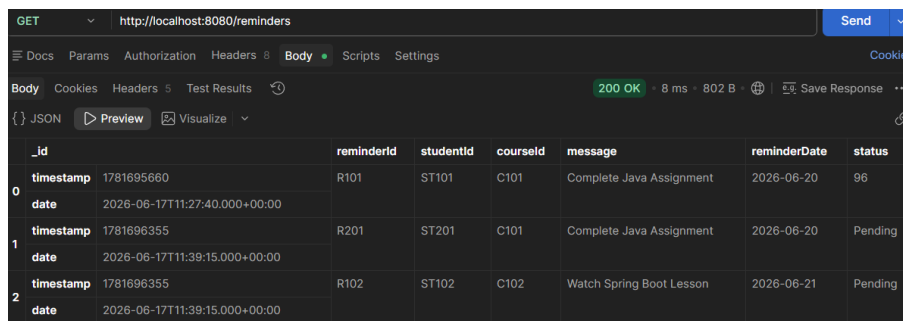
(b) Retrieve Reminder



(c) Get All Reminders



(d) Update Reminder



(e) All Reminder Records

Figure 9: Reminder Management Operations

13.10 Student Dashboard API

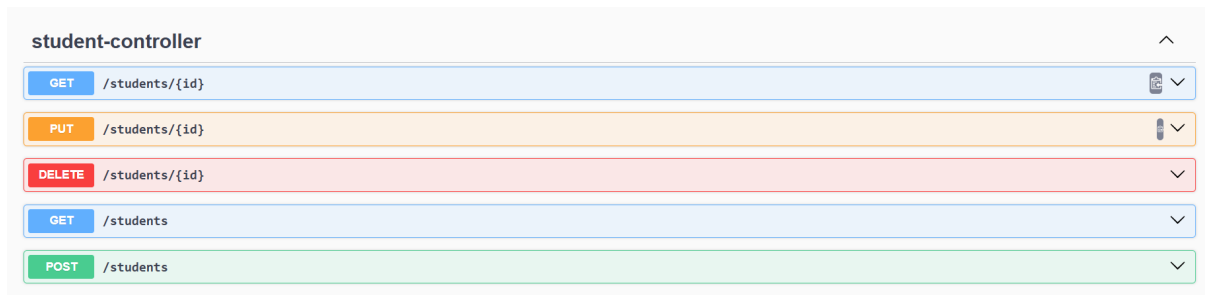
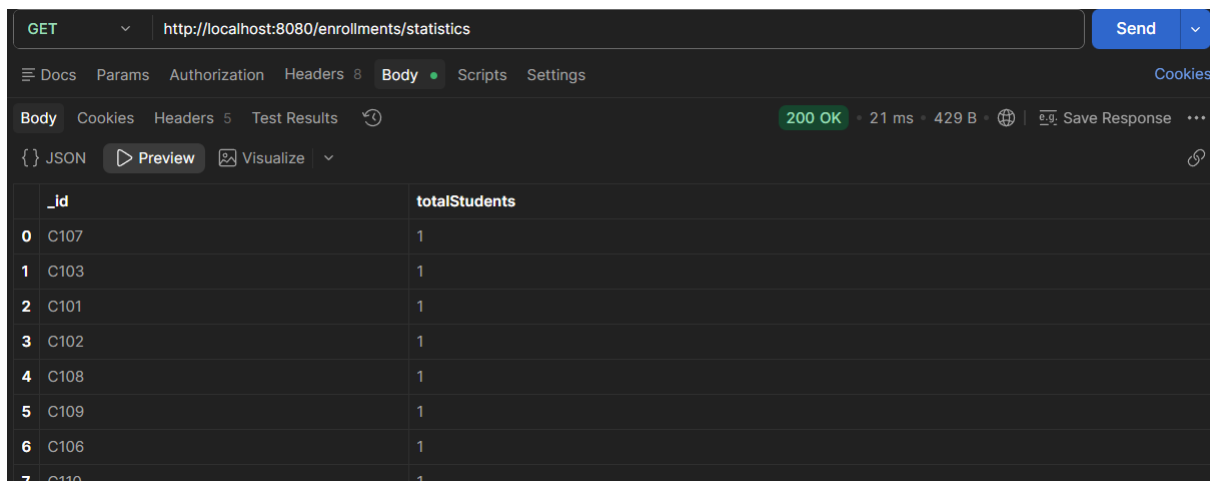


Figure 10: Student Dashboard API Using MongoDB Aggregation

13.11 Course Statistics API



A screenshot of an API client interface showing a GET request to `http://localhost:8080/enrollments/statistics`. The response is a 200 OK status with a 21 ms response time and 429 B of data. The response body is displayed in JSON format, showing a list of course statistics.

	_id	totalStudents
0	C107	1
1	C103	1
2	C101	1
3	C102	1
4	C108	1
5	C109	1
6	C106	1
7	C110	1

Figure 11: Course Statistics Generated Using Aggregation Framework

13.12 Course Completion Tracker

GET

http://localhost:8080/courses/completion-report

Send

Docs

Params

Authorization

Headers 8

Body

Scripts

Settings

Cookies

Body

Cookies

Headers 5

Test Results

200 OK

93 ms

2.14 KB

Save Response

JSON

Preview

Visualize

_id	averageCompletion	courseInfo							
0	C110 80	_id	timestamp	date	courseId	courseName	description	duration	instructorId
		0	1781442954	2026-06-14T13:15:54.000+00:00	C110	Machine Learning	ML Algorithms	60 Days	INS110
1	C107 90	_id	timestamp	date	courseId	courseName	description	duration	instructorId
		0	1781442954	2026-06-14T13:15:54.000+00:00	C107	AWS Cloud	Cloud Computing	45 Days	INS107
2	C102 60	_id	timestamp	date	courseId	courseName	description	duration	instructorId
		0	1781349052	2026-06-13T11:10:52.000+00:00	C102	MongoDB	MongoDB Fundamentals	20 Days	INS102
3	C101 63.75	_id	timestamp	date	courseId	courseName	description	duration	instructorId
		0	1781349052	2026-06-13T11:10:52.000+00:00	C101	Java Programming	Complete Java Course	30 Days	INS101
4	C103 90	_id	timestamp	date	courseId	courseName	description	duration	instructorId
		0	1781349052	2026-06-13T11:10:52.000+00:00	C103	Spring Boot	Spring Boot Development	25 Days	INS101
5	C108 60	_id	timestamp	date	courseId	courseName	description	duration	instructorId
		0	1781442954	2026-06-14T13:15:54.000+00:00	C108	DevOps	CI/CD and Docker	40 Days	INS108

Figure 12: Course Completion Tracking Using \$group and \$avg Aggregation

14 Testing

Testing Tool: Postman

Testing Performed:

- API Endpoint Testing
- CRUD Validation
- Aggregation Validation
- Relationship Validation

Result:

All APIs executed successfully and returned expected results.

15 Summary

The E-Learning Portal Backend System successfully implements a complete backend solution using Java Spring Boot and MongoDB. The project demonstrates CRUD operations, entity relationships, REST API development, MongoDB Aggregation Framework, reminder management, and reporting features.

The system provides a scalable architecture suitable for educational platforms and effectively showcases practical MongoDB concepts.

16 Conclusion

The project successfully achieved all objectives of creating an online learning management backend system. Through the implementation of MongoDB collections, relationships, aggregation pipelines, and REST APIs, the system provides efficient data management and reporting functionalities.

The project enhanced understanding of Java Spring Boot, MongoDB CRUD operations, aggregation framework, RESTful web services, API documentation, and backend application development.

Future enhancements may include authentication, certificates, notifications, reviews, payment integration, and real-time features for enterprise-level deployment.